1 Unions and Intersections

It is known that the elements of a set can be sets themselves. Rather than talking about a set of sets, it is common to use expressions such as a *family of sets* or a *collection of sets*.
As we will see later, a topology is a collection of sets which satisfy certain properties. It is therefore essential for us to be comfortable working with collections of sets, in particular their unions and intersections.

There are different ways to denote families of sets.

Notation 1 We will typically denote a family of sets using calligraphic letters such as $\mathcal{A}$, $\mathcal{B}$, $\mathcal{T}$, etc.

Definition 1. *Let $\mathcal{A}$ be a family of sets. We define*

$$\bigcup_{A\in\mathcal{A}} A = \{x \mid x \in A \text{ for some } A \in \mathcal{A}\} \quad \text{and} \quad \bigcap_{A\in\mathcal{A}} A = \{x \mid x \in A \text{ for all } A \in \mathcal{A}\}.$$

If the sets in a family are characterized by a simple condition, we may also use a notation for the union and intersection which displays explicitly the condition below the union or intersection symbol.

Example Let $\mathcal{A}$ be the collection of all subsets of $\mathbb{R}$ that do not contain 0, that is

$$\mathcal{A} = \{A \subseteq \mathbb{R} \mid 0 \notin A\}.$$

We may write the union of this collection of sets in any of the following ways

$$\bigcup_{A\in\mathcal{A}} A = \bigcup_{\substack{A\subseteq\mathbb{R}\\ 0\notin A}} A = \bigcup_{0\notin A\subseteq\mathbb{R}} A = \bigcup_{A\subseteq\mathbb{R}\setminus\{0\}} A$$

The last one follows from the observation that the subsets of $\mathbb{R}$ that do not contain 0 are exactly the subsets of $\mathbb{R}\setminus\{0\}$.
It should be intuitively clear that the union considered above is equal to the set of real numbers excluding zero, that is we have

$$\bigcup_{A\in\mathcal{A}} A = \mathbb{R}\setminus\{0\}.$$

Proof. We will show the double inclusion. First, let $x \in \bigcup_{A\in\mathcal{A}} A$. This means that $x \in A$ for some $A \in \mathcal{A}$, that is some $A \subseteq \mathbb{R}\setminus\{0\}$. Therefore, $x \in \mathbb{R}\setminus\{0\}$. This shows $\bigcup_{A\in\mathcal{A}} A \subseteq \mathbb{R}\setminus\{0\}$.
Conversely, let $x \in \mathbb{R}\setminus\{0\}$. Then $x \in \{x\}$ and $\{x\} \in \mathcal{A}$ since $\{x\}$ is a subset of $\mathbb{R}$ that does not contain 0. This shows $\mathbb{R}\setminus\{0\} \subseteq \bigcup_{A\in\mathcal{A}} A$, giving the other inclusion. $\square$

Notation 2 Another way to denote a family of sets is by using an indexing set I. This can be any set whose elements act as indices for the different sets of the family. More specifically, suppose that I is an indexing set and for each $i \in I$ we are given a set A_i. Then, we may denote the corresponding collection of sets as

$$\mathcal{A} = \{A_i \,|\, i \in I\} = \{A_i\}_{i \in I}.$$

In this notation, the union and intersection take the following form

$$\bigcup_{i \in I} A_i = \{x \mid x \in A_i \text{ for some } i \in I\} \quad \text{and} \quad \bigcap_{i \in I} A_i = \{x \mid x \in A_i \text{ for all } i \in I\}.$$

Special indexing sets have corresponding special notations. For example

$$\bigcup_{i=1}^{n} A_i \quad \text{can be used in place of} \quad \bigcup_{i \in \{1,2,\ldots,n\}} A_i$$

$$\bigcup_{i=1}^{\infty} A_i \quad \text{can be used in place of} \quad \bigcup_{i \in \mathbb{N}} A_i$$

Similar conventions hold for intersections.

Example We claim that

$$\bigcup_{n \in \mathbb{N}} [n^{-1}, 2] =]0, 2].$$

Proof. We will show the double inclusion. First, let $x \in \bigcup_{n \in \mathbb{N}} [n^{-1}, 2]$. This means that $x \in [n^{-1}, 2]$ for some $n \in \mathbb{N}$. Since $n^{-1} > 0$, we get $x \in]0, 2]$. This shows $\bigcup_{n \in \mathbb{N}} [n^{-1}, 2] \subseteq]0, 2]$.

Next, let $x \in]0, 2]$. By the Archimidean principle (or the definition of limit applied to $n^{-1} \to 0$) there exists $n \in \mathbb{N}$ such that $n^{-1} \leq x$, thus $x \in [n^{-1}, 2]$. This shows $x \in \bigcup_{n \in \mathbb{N}} [n^{-1}, 2]$ and gives the other inclusion. $\square$

2 Images and Preimages

Definition 2. *Let $f : X \to Y$ be a function.*

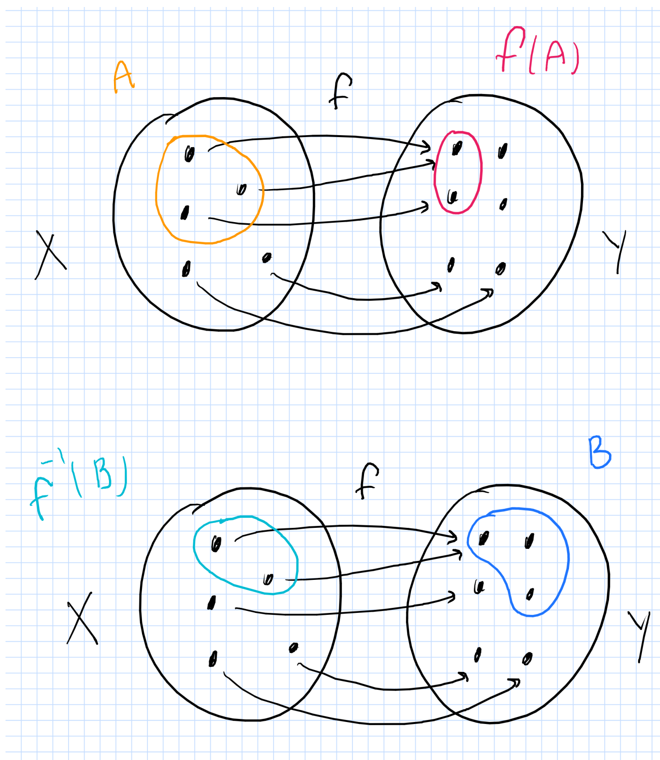
- *If $B \subseteq Y$, the* **preimage** *of B under f is*

$$f^{-1}(B) = \{x \in X \mid f(x) \in B\}$$

- *If $A \subseteq X$, the* **image of** A **under** f *is*

$$f(A) = \{f(x) \in Y \mid x \in A\} = \{y \in Y \mid \exists x \in A \ \ s.t. \ \ y = f(x)\}$$

- *The* **image of** f *(or* **range of** f*) is the image of the domain X under f, and it may be denoted by $f(X)$, $Im(f)$, or $Range(f)$.*



3 Important Results

Theorem 1 (De Morgan's Laws). *Let $\{A_i\}_{i\in I}$ be a family of sets. Then*

a) $$\left(\bigcup_{i\in I} A_i\right)^c = \bigcap_{i\in I} A_i^c$$ *(the complement of the union is the intersection of the complements)*

b) $$\left(\bigcap_{i\in I} A_i\right)^c = \bigcup_{i\in I} A_i^c$$ *(the complement of the intersection is the union of the complements)*

where the complements are taken with respect to some universal set.

Proposition 1. *Let $f : X \longrightarrow Y$ be a function and let $\{A_i\}_{i\in I}$ be a family of subsets of X and $\{B_j\}_{j\in J}$ a family of subsets of Y . Then*

i) $$f\left(\bigcup_{i\in I} A_i\right) = \bigcup_{i\in I} f(A_i)$$ *(the image of the union is the union of the images)*

ii) $$f\left(\bigcap_{i\in I} A_i\right) \subseteq \bigcap_{i\in I} f(A_i)$$ *(the image of the intersection is the intersection of the images)*

iii) $$f^{-1}\left(\bigcup_{j\in J} B_j\right) = \bigcup_{j\in J} f^{-1}(B_j)$$ *(the preimage of the union is the union of the preimages)*

iv) $$f^{-1}\left(\bigcap_{j\in J} B_j\right) = \bigcap_{j\in J} f^{-1}(B_j)$$ *(the preimage of the intersection is the intersection of the preimages)*

The following result is often used in Topology without even being mentioned. I am highlighting it as a standalone proposition. We will revisit this in class a number of times under different points of view, but I recommend you spend some time now trying to really understand it.

Proposition 2. *Let $\mathcal{A}$ be a collection of sets. A set Z can be written as a union of elements of $\mathcal{A}$ if and only if for every $x \in Z$ there exists $A_x \in \mathcal{A}$ such that $x \in A_x \subseteq Z$.*

4 Important Exercises

Exercise 1 Find the following unions and intersections.

i) $\bigcup_{n\in\mathbb{N}}[0, 2 - n^{-1}]$

ii) $\bigcap_{\varepsilon>0}[1, 1+\varepsilon]$

Exercise 2 Prove Proposition 2

5 Additional Practice

Exercise 3 Find the following unions and intersections.

i) $\bigcup_{x\in\mathbb{R}}\{x^2\}$

ii) $\bigcup_{A\subseteq\mathbb{Q}} A$

iii) $\bigcap_{B\in\mathcal{B}} B$ where $\mathcal{B}$ is the family of all intervals in $\mathbb{R}$ containing the origin

Exercise 4 Prove Theorem 1
Exercise 5 Prove Proposition 1